Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Linköpings universitet
Short name / acronym used in projects	LiU
Participant Identification Code (PIC), if available	999852236
Website(s)	www.liu.se https://liu.se/en/employee/neina50 https://liu.se/en/research/cornea-research
Country	Sweden
Organisation type	x Academic ☐ Non-academic x Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Campus Valla, Linköping, 581 83
Department(s), institute(s), research group(s) or unit(s) that would be involved	Division of Ophthalmology, Department of Biomedical and Clinical Sciences
Additional institutional resources relevant to collaboration	Full core facility including omics, molecular biology, sequencing, microscopy, animal care and housing, access to supercomputer for computational work, full ophthalmology research facility including molecular biology and cell culture labs, biobank, femtosecond laser surgical suite for preclinical research, in vivo diagnostic and imaging equipment for preclinical research, full bioinformatics expertise

Main contact person for this profile	Neil Lagali, Professor, PI of eye diseases research group, Division of Ophthalmology, Dept of Biomedical and Clinical Sciences
Main contact email	neil.lagali@liu.se
Other contact person(s), if relevant	Petros Moustardas, Research Associate, Co-supervisor and mentor, research and technical support: petros.moustardas@liu.se

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
X Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	X Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	X Optometry / visual behaviour / rehabilitation
X Advanced therapies / regenerative medicine	X Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	X Drug delivery / formulations / nanomedicine
X Drug discovery / screening / repurposing	X Ocular vascular disease
X Infection / trauma-related ocular disease	X Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	X AI-assisted diagnosis / software / digital tools
X Clinical trials / translational validation	X Other (please specify): ocular imaging, ocular omics, rare ocular diseases, ocular medical devices

Free Keywords for internal searchable database (please provide up to 10)	Ocular surface disease, limbal stem cells, rare eye diseases, AI-based decision support, corneal regenerative medicine, drug repurposing, transcriptomics, proteomics, tear film, neuropathic ocular pain
Geographies of interest for exchanges (optional)	all

Overview of your research / innovation focus
The ocular research group at Linköping University has the following research focuses: nAMD: Longitudinal OCTA studies of nAMD patients receiving anti-VEGF injections, including proteome profiling (plasma and tears) and miRNome profiling long-COVID studies: eye symptoms in long-COVID, tests of ocular and autonomic nerve/brain function, neuroinflammatory pathways and mechanisms, tear proteome profiling, visual training and rehabilitation Corneal nerves and neuropathic pain/allodynia: corneal nerve imaging in patients, corneal nerve regeneration studies with experimental therapies (preclinical models), experimental treatment of neuropathic pain (clinical studies) Limbal stem cell deficiency: preclinical models, scRNA-seq and spatial transcriptomics analyses in mouse models, evaluation of experimental therapies (drug repurposing), PAX6 mutations and role in corneal development, homeostasis and development of aniridia-associated keratopathy, effects of p63 mutations including preclinical model of AEC-EEC syndrome Corneal angiogenesis: preclinical studies of mechanisms of angiogenesis/neovascularization in the cornea including experimental therapy evaluation

Diabetes studies: ocular impact of type 2 diabetes including corneal nerves, diabetic retinopathy, long-term longitudinal corneal imaging studies in diabetes cohorts, scRNA-seq of blood, proteomics of blood and tears

Corneal regeneration and biomaterials: evaluation of new biomaterials and medical devices for implantation in the cornea to treat degenerative and refractive disorders, including laboratory validation, ex vivo studies, development of novel surgical methods (femtosecond surgical laser), preclinical large-animal models, clinical translation

Deep learning and corneal imaging: collaborative work with confocal microscopy images and their use in disease diagnosis, deep-learning assisted pipelines for image processing and analysis, clinical decision support

Infectious keratitis: epidemiological studies of various IK, including clinical examinations, retrospective analyses, evaluation of treatment, diagnostics, etc.

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

Our research team develops new therapies, devices, treatments and diagnostics from a basic science research level through to preclinical validation and into first human clinical trials. We also have projects exclusively focused on mechanisms of disease and while others are clinically-focused on improving understanding and therapy of eye diseases. We have a broad area of interest and deep expertise in cornea and ocular surface disease models and mechanisms, experimental therapy evaluation and imaging. We bring a long track record of ocular surface disease research to the consortium including various research tools, methods and platforms.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.

- transgenic and acquired mouse models of limbal stem cell deficiency
- mouse models of ocular inflammation, neurodegeneration, neovascularization
- rabbit models of corneal inflammation, neurotrophic keratopathy, neovascularization
- ocular drug delivery approaches
- biomaterials use in the eye, slow release drug delivery
- experimental femtosecond laser surgery
- ocular imaging, in vivo diagnostics, corneal biomechanics
- transcriptomics (bulk, single cell, spatial) of cornea tissue, blood, proteomics of blood and tear film
- bioinformatics analysis and data visualization for large datasets
- in vivo confocal microscopy imaging, image processing, AI-based analysis and diagnostic tools
- translational research and surgery facility

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

- full basic science and translational ophthalmology labs including cell culture lab
- core facility infrastructures (sequencing, microscopy, animal work, omics)
- bioinformatics expertise

- femtosecond laser, biomechanical analyzer, OCT, in vivo confocal microscopy, meibomography, topography, dynamic pupillometry, etc.
- corneal image datasets (published and non-published)
- omics datasets (some analyzed, others un-analyzed)
- facilities to host researchers (office space, IT support, lab space, etc.)
- biobank of patient-derived blood and tear samples

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

- scientific skills in various molecular biology and translational workflows
-data analysis and processing using existing datasets including bioinformatics
- observation of clinical data collection, imaging
- learning image analysis and AI-based diagnostic tools
- skills in cell and biomaterial laboratory work
- collaborative research, performing laboratory work and analyses as part of collaborative research projects

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

- new collaborations to move forward translational research towards clinical use
  - we can provide expertise and methods, resources, etc., that extend beyond ophthalmology and are applicable to molecular biology generally
  - we have projects involving COVID-19, type 2 diabetes, Parkinson's disease, etc.
  - we have experience in designing and conducting clinical studies and trials, including ethics, protocol and documentation requirements

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Neil Lagali	PI	Ophthalmology	Professor	senior	neil.lagali@liu.se
Petros Moustardas	Supervision, training	Ophthalmology	Research Associate	Mid-stage	petros.moustardas@liu.se
Ava Dashti	Collaboration, training, supervision	Ophthalmology	Doctoral candidate	Early-stage	ava.dashti@liu.se

Main disciplines represented in your team	Ophthalmology, optometry, cell biology, genetics, omics, biomaterials, AI, medical devices, imaging, biomarkers, clinical trials, preclinical models.
Experience in EU / international / multi-partner collaborative projects	Current partner in Horizon Europe project consortium, past coordinator of 2 EU project consortia, past MC member of 2 EU COST Actions (1 as Action Chair), current member of 2 EU COST Actions, past member of 1 Marie Curie RISE consortium, current member of 1 Marie Curie Doctoral Network
Experience with clinical, translational or patient-facing research (if relevant)	X Yes ☐ No ☐ Not applicable | Short note: clinical observational and interventional studies, translation from bench to clinic

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

Romano D, Romano V, Daruich A, Ferrari G, Robert M, Semeraro F, Lagali N, Bremond-Gignac D; ANIRIDIA-NET Clinical Guidelines Working Group; Aniridia-Europa and Geniris Working Group. Congenital aniridia: European COST action ANIRIDIA-NET guidelines for diagnosis, management and care. Acta Ophthalmol. 2026 May;104(3):267-279. doi: 10.1111/aos.17587. Epub 2025 Sep 2. PMID: 40892309; PMCID: PMC13058686.

Dina Javidjam, Meri Vattulainen, Neil Lagali, Petros Moustardas. Decoding murine corneal epithelial specification and homeostasis by single-cell spatial transcriptomics with scRNA-seq enrichment.
bioRxiv 2026.05.08.723186; doi: https://doi.org/10.64898/2026.05.08.723186

Ji M, Song Y, Roth J, Dashti A, Lazo J, Lincke A, Macedo AFT, Löwe W, Lagali N. Deep learning-based segmentation and density estimation of corneal nerves and dendritic cells from In Vivo confocal microscopy images. Sci Rep. 2026 Jan 13;16(1):1620. doi: 10.1038/s41598-025-34412-6. PMID: 41530230; PMCID: PMC12800303.

Javidjam D, Moustardas P, Abbasi M, Dashti A, Rautavaara Y, Lagali N. A human-like model of aniridia-associated keratopathy for mechanistic and therapeutic studies. JCI Insight. 2024 Dec 3;10(2):e183965. doi: 10.1172/jci.insight.183965. PMID: 39625791; PMCID: PMC11790027.

Moustardas P, Abbasi M, Javidjam D, Asamoah CS, Schweitzer-Chaput A, Cisternino S, Bremond-Gignac D, Aberdam D, Lagali N. Duloxetine enhances PAX6 expression and suppresses innate immune responses in murine LPS-induced corneal inflammation. Ocul Surf. 2024 Oct;34:225-234. doi: 10.1016/j.jtos.2024.08.008. Epub 2024 Aug 8. PMID: 39127390.

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). ***Indicate if international.***

Stem Cell Therapy Development for Corneal Epithelium Regeneration, STEM-CORE, Marie Curie Doctoral Network, 101226474, 2025-2028. Role: Beneficiary (International)

Novel advanced and repurposed therapeutics for vision restoration in a group of severe rare ocular surface diseases: from validation to first clinical investigations, RESTORE VISION, EU Horizon Europe, 101080611, 2024-2027, Role: Beneficiary (International)

Networking to address an unmet medical, scientific, and societal challenges, ANIRIDIA-NET, EU COST Action CA18116, European Union COST Programme, 2019-2023, Role: Coordinator/Action Chair (International)

Aniridia – Novel therapeutic tools to treat or prevent progressive cornea opacification, AAK-INSIGHT, European Joint Programme on Rare Diseases (EJPRD), EJPRD20-135, 2020-2024, Role: Coordinator (International)

Advanced Regenerative and RESTtorative Therapies to combat corneal BLINDNESS, ARREST BLINDNESS, EU Horizon2020, 667400, 2016-2020, Role: Coordinator (International)

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Please indicate only what can be shared at this stage.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers x Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff x Other profiles: clinical specialists and residents
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Insurance from home country needed

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile. cornea, ocular surface disease, regenerative medicine, omics, stem cells, imaging

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. - complementary expertise, whatever makes sense

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
A form is needed to be signed by the division head prior to receiving guest researchers. Guests will undergo lab safety training.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

x Understood

Name: Neil Lagali
Date completed: May 29, 2026

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